

UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing
SEMESTER 1 SESI 2024 / 2025

LAB ACTIVITY

SECP 2523 DATABASE (WBL)

AHLI KUMPULAN :

- | | | |
|----|------------------------------------|------------------|
| 1. | <u>LAU YAN KAI</u> | <u>A23CS0098</u> |
| 2. | <u>CHEW CHIU XIAN</u> | <u>A23CS0061</u> |
| 3. | <u>AHMAD ZIYAAD BIN MOHD ABBAS</u> | <u>A23CS0206</u> |

LAB 1 - Creating a Database

CREATING THE TENNIS LOGOS DATABASE

Tennis Logos (TL) supplies customized tennis clothing and accessories for clubs and tournaments. The company purchases these items from suppliers at wholesale prices and adds the customer's logo. The final item price is determined by marking up the wholesale price and adding a fee that is based on the complexity of the logo.

Tennis Logos also does graphic design work for customers. Currently, the information about the items and the suppliers is stored in two Excel workbooks. Each item is assigned to a single supplier, but each supplier may be assigned many items.

You are to create a database that will store the item and supplier information. You already have determined that you need two tables, a Supplier table and an Item table, in which to store the information.

Instructions: Perform the following tasks.

1. Use the Blank desktop database option to create a new database in which to store all objects related to the items for sale. Call the database – TL_Sport Database.

Answer:

```
mysql> CREATE DATABASE TL_Sport;  
Query OK, 1 row affected (0.01 sec)  
  
mysql> USE TL_Sport;  
Database changed
```

2. Create the Supplier table using the structure shown in Figure 1. Use the name Supplier for the table.

Answer:

```
mysql> CREATE TABLE Supplier(
    -> SupplierCode CHAR(2) PRIMARY KEY,
    -> SupplierName VARCHAR(30),
    -> TelephoneNum VARCHAR(12)
    -> );
Query OK, 0 rows affected (0.59 sec)
```

3. Add the data shown in Figure 2 to the Supplier table.

Answer:

```
mysql> INSERT INTO Supplier (SupplierCode, SupplierName, TelephoneNum)
    -> VALUES
    -> ('LM', 'Last Merchandisers', '803-555-7641'),
    -> ('PC', 'Pratt Clothing', '517-555-3853'),
    -> ('SD', 'Scryps Distributors', '610-555-8741')
    -> ;
Query OK, 3 rows affected (0.23 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

4. Print the Supplier table.

Answer:

```
mysql> DESCRIBE Supplier;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| SupplierCode   | char(2)       | NO   | PRI | NULL    |       |
| SupplierName   | varchar(30)   | YES  |     | NULL    |       |
| TelephoneNum   | varchar(12)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.11 sec)

mysql> SELECT * FROM Supplier;
+-----+-----+-----+
| SupplierCode | SupplierName      | TelephoneNum |
+-----+-----+-----+
| LM           | Last Merchandisers | 803-555-7641 |
| PC           | Pratt Clothing     | 517-555-3853 |
| SD           | Scryps Distributors | 610-555-8741 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

5. Create the Item table using the structure shown in Figure 3. Use the name Item for the table.

Answer:

```
mysql> CREATE TABLE Item(
  -> ItemNo CHAR(4) PRIMARY KEY,
  -> Description VARCHAR(20),
  -> OnHand INT,
  -> WholesalePrice DECIMAL(10,2),
  -> BaseCost DECIMAL(10,2),
  -> SupplierCode CHAR(2),
  -> FOREIGN KEY (SupplierCode) REFERENCES Supplier(SupplierCode)
  -> );
Query OK, 0 rows affected (0.41 sec)
```

6. Add the data shown in Figure 4 to the Item table.

Answer:

```
mysql> INSERT INTO Item (ItemNo, Description, OnHand, WholesalePrice, BaseCost, SupplierCode)
  -> VALUES
  -> ('3363', 'Baseball Cap', 110, 4.87, 6.10, 'LM'),
  -> ('3673', 'Cotton Visor', 150, 4.59, 5.74, 'LM'),
  -> ('4543', 'Crew Sweatshirt', 75, 7.29, 8.75, 'PC'),
  -> ('4583', 'Crew T-Shirt', 135, 2.81, 3.38, 'PC'),
  -> ('5923', 'Drink Holder', 75, 0.82, 1.07, 'SD'),
  -> ('6185', 'Fleece Vest', 50, 28.80, 34.50, 'PC'),
  -> ('6234', 'Foam Visor', 200, 0.79, 1.00, 'LM'),
  -> ('6345', 'Golf Shirt', 115, 10.06, 12.35, 'PC'),
  -> ('7123', 'Sports Bottle', 150, 1.04, 1.35, 'SD'),
  -> ('7934', 'Tote Bag', 225, 1.26, 1.75, 'SD'),
  -> ('8136', 'Travel Mug', 80, 2.47, 3.50, 'SD'),
  -> ('8344', 'V-Neck Pullover', 76, 13.60, 15.75, 'LM'),
  -> ('8590', 'V-Neck Vest', 65, 20.85, 25.02, 'LM'),
  -> ('9458', 'Windbreaker', 54, 15.17, 18.20, 'PC'),
  -> ('9620', 'Zip Hoodie', 90, 16.67, 20.04, 'PC');
Query OK, 15 rows affected (0.13 sec)
Records: 15 Duplicates: 0 Warnings: 0
```

7. *Print the Item table.*

Answer:

```
mysql> DESCRIBE Item;
```

Field	Type	Null	Key	Default	Extra
ItemNo	char(4)	NO	PRI	NULL	
Description	varchar(20)	YES		NULL	
OnHand	int	YES		NULL	
WholesalePrice	decimal(10,2)	YES		NULL	
BaseCost	decimal(10,2)	YES		NULL	
SupplierCode	char(2)	YES	MUL	NULL	

```
6 rows in set (0.04 sec)
```

```
mysql> SELECT * FROM Item;
```

ItemNo	Description	OnHand	WholesalePrice	BaseCost	SupplierCode
3363	Baseball Cap	110	4.87	6.10	LM
3673	Cotton Visor	150	4.59	5.74	LM
4543	Crew Sweatshirt	75	7.29	8.75	PC
4583	Crew T-Shirt	135	2.81	3.38	PC
5923	Drink Holder	75	0.82	1.07	SD
6185	Fleece Vest	50	28.80	34.50	PC
6234	Foam Visor	200	0.79	1.00	LM
6345	Golf Shirt	115	10.06	12.35	PC
7123	Sports Bottle	150	1.04	1.35	SD
7934	Tote Bag	225	1.26	1.75	SD
8136	Travel Mug	80	2.47	3.50	SD
8344	V-Neck Pullover	76	13.60	15.75	LM
8590	V-Neck Vest	65	20.85	25.02	LM
9458	Windbreaker	54	15.17	18.20	PC
9620	Zip Hoodie	90	16.67	20.04	PC

```
15 rows in set (0.00 sec)
```

8. *Create a query for the Item table. Include the Item Number, Description, Wholesale Price, Base Cost, and Supplier Code. Save the query as Item Query.*

Answer:

Result Grid					
Filter Rows:					
Edit:					
	ItemNo	Description	WholesalePrice	BaseCost	SupplierCode
▶	3363	Baseball Cap	4.87	6.10	LM
	3673	Cotton Visor	4.59	5.74	LM
	4543	Crew Sweatshirt	7.29	8.75	PC
	4583	Crew T-Shirt	2.81	3.38	PC
	5923	Drink Holder	0.82	1.07	SD
	6185	Fleece Vest	28.80	34.50	PC
	6234	Foam Visor	0.79	1.00	LM
	6345	Golf Shirt	10.06	12.35	PC
	7123	Sports Bottle	1.04	1.35	SD
	7934	Tote Bag	1.26	1.75	SD
	8136	Travel Mug	2.47	3.50	SD
	8344	V-Neck Pullover	13.60	15.75	LM
	8590	V-Neck Vest	20.85	25.02	LM
	9458	Windbreaker	15.17	18.20	PC
	9620	Zip Hoodie	16.67	20.04	PC
•	NULL	NULL	NULL	NULL	NULL

9. Create a simple form for the *Item* table. Use the name *Item Form* for the form.

Answer:

10. *Drop the Supplier table.*

Answer:

```
mysql> ALTER TABLE Item DROP FOREIGN KEY item_ibfk_1;  
Query OK, 0 rows affected (0.07 sec)  
Records: 0  Duplicates: 0  Warnings: 0  
  
mysql> DROP TABLE Supplier;  
Query OK, 0 rows affected (0.04 sec)
```


Structure of Supplier Table

FIELD NAME	DATA TYPE	FIELD SIZE	PRIMARY KEY?	DESCRIPTION
Supplier Code	Short Text	2	Yes	Supplier Code (Primary Key)
Supplier Name	Short Text	30		Supplier Name
Telephone Number	Short Text	12		Supplier Telephone Number

Figure 1

Data for Supplier Table

Supplier Code	Supplier Name	Telephone Number
LM	Last Merchandisers	803-555-7641
PC	Pratt Clothing	517-555-3853
SD	Scryps Distributors	610-555-8741

Figure 2

Structure of Item Table

FIELD NAME	DATA TYPE	FIELD SIZE	PRIMARY KEY?	DESCRIPTION
Item Number	Short Text	4	Yes	Item Number (Primary Key)
Description	Short Text	20		Description of Item
On Hand	Number			On-Hand Item
Wholesale Price	Currency			Wholesale Price (Caption)
Base Cost	Currency			Base Cost for Item
Supplier Code	Short Text	2		Supplier Code

Figure 3

Data for Item Table

Item Number	Description	On Hand	Wholesale Price	Base Cost	Supplier Code
3363	Baseball Cap	110	\$4.87	\$6.10	LM
3673	Cotton Visor	150	\$4.59	\$5.74	LM
4543	Crew Sweatshirt	75	\$7.29	\$8.75	PC
4583	Crew T-Shirt	135	\$2.81	\$3.38	PC
5923	Drink Holder	75	\$0.82	\$1.07	SD
6185	Fleece Vest	50	\$28.80	\$34.50	PC
6234	Foam Visor	200	\$0.79	\$1.00	LM
6345	Golf Shirt	115	\$10.06	\$12.35	PC
7123	Sports Bottle	150	\$1.04	\$1.35	SD
7934	Tote Bag	225	\$1.26	\$1.75	SD
8136	Travel Mug	80	\$2.47	\$3.50	SD
8344	V-Neck Pullover	76	\$13.60	\$15.75	LM
8590	V-Neck Vest	65	\$20.85	\$25.02	LM
9458	Windbreaker	54	\$15.17	\$18.20	PC
9620	Zip Hoodie	90	\$16.67	\$20.04	PC

Figure 4